

A disproof of Conjecture 00000000748

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Statement and counterexample

The conjecture classifies the polynomials compatible with the Teichmüller lift $w : \mathbb{F}_p \rightarrow \mathbb{Z}_p$, claiming that the only minimum-degree examples satisfying $P \circ w = w \circ \overline{P}$ are identity and Frobenius. As written, constant polynomials are not excluded.

Theorem 1. *The stated classification is false.*

Proof. Take $P(X) = 0 \in \mathbb{Z}_p[X]$ and $\overline{P}(X) = 0 \in \mathbb{F}_p[X]$. The defining property $w(0) = 0$ gives, for every $x \in \mathbb{F}_p$,

$$P(w(x)) = 0 = w(0) = w(\overline{P}(x)).$$

Hence the required commuting identity holds. The polynomial has minimum possible degree (degree zero among nonempty coefficient lists; under the standard convention the zero polynomial has degree $-\infty$). Nevertheless it is neither identity nor Frobenius: at 1 both listed maps have value 1, whereas $P(1) = 0$. $\square$

The same argument works with the constant-one polynomial. Therefore any valid classification must at least add a nonconstant or normalization hypothesis; neither occurs in the filed conjecture.

Lean 4 certificate

The project in `lean4/` defines the commuting-square predicate `LiftCompatible`. The theorem `counterexample` constructs the zero maps, proves compatibility from $w(0) = 0$, and proves by evaluation at 1 that the map is neither identity nor Frobenius. It contains no `sorry`, `native_decide`, or additional axiom.